MemoLens

Simulated smart glasses for recognizing familiar people, daily reminders, and an AI chatbot, built on Microsoft Azure

Project Documentation | Microsoft Machine Learning Track | Digital Egypt Pioneers Initiative (DEPI), Phase Five

Project name	MemoLens
Track	Microsoft Machine Learning Track
Cloud platform	Microsoft Azure
Project type	Educational assistive prototype (graduation project)
Platform	Web application that simulates smart glasses on a phone or laptop, using the device's existing camera
GitHub repository	https://github.com/nayra-ahmedaraby/MemoLens
Team leader	Nayra Ahmed
Submission date	19 October 2026

Contents

- **Part 1 – Project Planning & Management:** idea, objectives, scope, architecture, timeline, team responsibilities, risks, and success metrics.
- **Part 2 – Literature Review:** similar solutions, their strengths and limitations, comparison with MemoLens, and lessons applied.
- **Part 3 – Requirements Gathering:** stakeholders and their needs, functional requirements, and quality (non-functional) requirements.
- **References**

MemoLens is an educational assistive prototype. It is not a medical device, does not diagnose, treat, or prevent Alzheimer's disease, and is not a replacement for professional care or caregivers.

PART 1

Project Planning & Management

Idea, objectives, scope, plan, team, risks, and success metrics

1.1 Project Idea

People living with Alzheimer's disease often fail to recognize family members and friends, forget appointments and medication, and ask the same personal questions repeatedly. MemoLens is an assistive system that supports them in these daily situations. The long-term vision is a pair of camera-equipped smart glasses that recognize familiar people and remind the wearer of their daily schedule and medication, together with an AI chatbot that answers personal questions from trusted information.

In this project the glasses are **simulated**: a web application running on a phone or laptop uses the device's camera as the glasses' camera and shows what the wearer would see and hear. An authorized family member or caregiver manages all familiar-person profiles, approved facts, and schedules; the patient-facing view presents only this approved content. The system is designed, trained, deployed, and monitored on **Microsoft Azure**.

1.2 Objectives

- Train and deploy a face-recognition model in Azure Machine Learning that identifies caregiver-registered people and reports unknown or uncertain faces instead of guessing.
- Deliver daily schedule and medication reminders to the patient at the scheduled time.
- Provide an AI chatbot that answers the patient's questions using caregiver-approved information only.
- Give caregivers a secure interface to manage profiles, reference images, facts, and reminders.
- Apply Microsoft's Responsible AI principles and protect personal data throughout the system.
- Deliver an integrated MVP deployed on Azure by Week 7 and a tested final system by Week 10.

1.3 Project Scope

The project includes three features:

Feature	Description
Face Recognition (Glasses Simulation)	Simulates the smart glasses' camera using the phone or laptop camera. Recognizes familiar people registered by the caregiver and shows their name and relationship on screen, as the glasses would. The model is trained, registered, and deployed with Azure Machine Learning. Unknown or uncertain faces are reported as such rather than guessed.
Daily Schedule & Medication Reminders	Displays the daily schedule and sends reminders triggered by Azure Functions. The system does not verify whether medication was actually taken.
AI Chatbot	A text chatbot built with Azure OpenAI that answers the patient's questions (for example, "Who is my daughter?" or "What is my next appointment?") using caregiver-approved information only. If information is missing, it says so instead of inventing an answer.

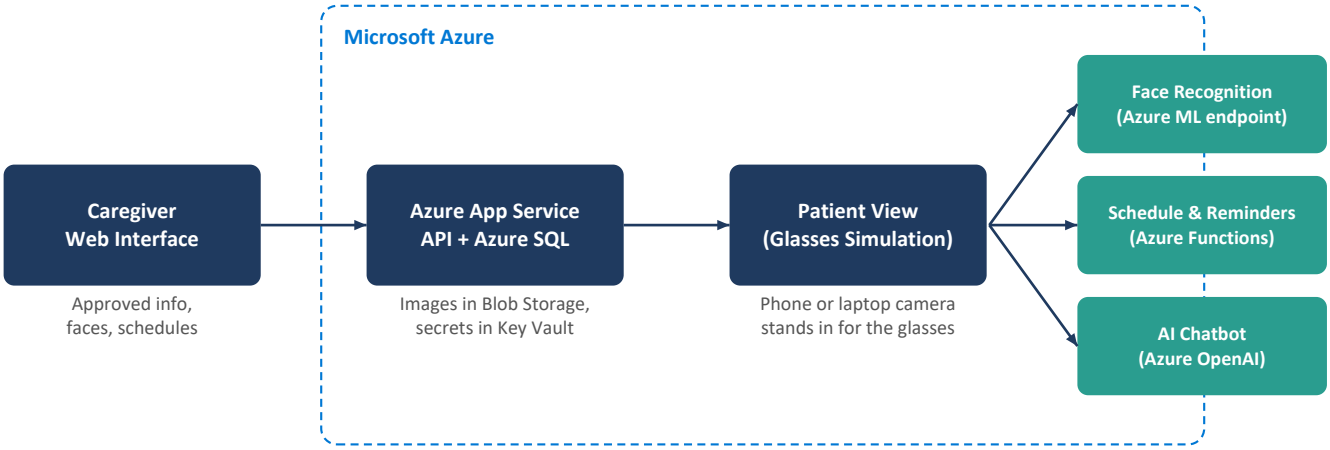
- **Platform:** A web application, hosted on Azure, that runs on a phone or laptop and uses its built-in camera to simulate the glasses.
- **Hardware:** No hardware is built in this project. Physical smart glasses are future work.
- **Future work:** Physical smart glasses, a voice assistant for hands-free questions and announcements, and a fuller trusted-information assistant.
- **Out of scope:** Fall detection, medical diagnosis, treatment advice, and proof of medication intake.
- **Status:** Planning / early development. No feature has been implemented yet and no results are reported; Azure tiers and model choices remain subject to feasibility testing.

1.4 Microsoft Azure Technology Stack

Component	Azure service	Purpose
Face-recognition model	Azure Machine Learning (training, MLflow, model registry, managed online endpoint)	Train and evaluate FaceNet / ArcFace embeddings and serve real-time face matching.
AI chatbot	Azure OpenAI Service + Azure AI Content Safety	Answer from caregiver-approved records only; filter unsafe content.
Reminders	Azure Functions (timer triggers)	Check schedules and send browser push notifications.
Backend & data	Azure App Service (FastAPI), Azure SQL Database, Blob Storage	Host the API and web views; store profiles, facts, schedules, and private reference images.
Security & operations	Microsoft Entra ID, Key Vault, Application Insights, GitHub Actions	Caregiver sign-in, secret storage, monitoring, and automated deployment.

Note: the Azure AI Face service's identification features are Limited Access and require Microsoft approval. The team therefore plans to train and host its own face-recognition model in Azure Machine Learning, and to use Azure AI Face only if access is granted.

1.5 System Workflow



The patient uses the glasses-simulation view to see who is in front of them, receive daily reminders, and ask the chatbot questions. The caregiver creates and updates profiles (with consented reference images), approved facts, and reminders. All data and models stay inside the team's Azure subscription.

1.6 Timeline

This is a proposed timeline, not an official DEPI schedule.

Week	Activities	Deliverable
Week 1	Confirm scope and requirements, create the GitHub repository and Azure resource group, assign responsibilities.	Repository, task board
Week 2	Design the user interface, database, API, and Azure architecture; set up the Azure ML workspace.	Design documents
Week 3	Prepare datasets and module prototypes; run first training experiments in Azure ML.	First model baseline
Week 4	Implement face recognition, reminders, the chatbot, and the glasses-simulation view in parallel.	Working modules
Week 5	Continue implementation, deploy the model endpoint, and test individual modules.	Deployed endpoint

Week	Activities	Deliverable
Week 6	Integrate the three features on Azure.	Integrated build
Week 7	Deliver an integrated MVP deployed on Azure.	MVP
Week 8	Test the complete system, review the Responsible AI dashboard, and fix errors.	Test report
Week 9	Improve the interface; prepare documentation, README, and presentation slides.	Final documentation
Week 10	Final review, demo preparation, and submission.	Final demo

1.7 Team Members and Responsibilities

All members test their own modules and contribute to integration on Azure and the final presentation.

#	Member	Role	Responsibilities
1	Nayra Ahmed	Team Leader & Integration	Coordinate tasks and deadlines, manage GitHub, documentation, and the Azure subscription and resource groups, track requirements and blockers, support API integration, and coordinate integration testing.
2	Nada Kotkat	Face Recognition	Develop, train, and test the face-recognition model in Azure Machine Learning and deploy it to a managed endpoint.
3	Mariam Mohamed	AI Chatbot	Develop the chatbot with Azure OpenAI using caregiver-approved information, and support face-recognition data preparation and testing.
4	Michael Samaan	Reminders & Daily Schedule	Develop the scheduling and reminder module with Azure Functions.
5	Naheel Mohamed	Backend & Database	Handle backend APIs on Azure App Service, Azure SQL database design, Blob Storage, and data management.
6	Mahmoud Elsoly	Frontend	Build the caregiver interface and the glasses-simulation view, connect them to the backend, and deploy on Azure App Service.

1.8 Risks and Mitigation

Risk	Impact	Mitigation
False face identification	High	Conservative threshold, unknown-face handling, testing under varied lighting and angles, evaluation in Azure ML.
Simulation differs from real glasses	Medium	Document the differences (camera angle, field of view, processing power); treat hardware as future work.
Chatbot invents answers	High	Approved-information-only answers with a fallback for missing facts; content filtering.
Reminders fail in background or offline	Medium	Test browser and device states; document limitations.
Exposure of personal data	High	Consent, Entra ID access control, private Blob Storage, Key Vault, synthetic data first, no secrets in the repository.
Azure credits run out or access is denied	Medium	Budget alerts, shut down idle compute, self-hosted model instead of Limited Access services.
Scope exceeds available time	Medium	Fixed three-feature scope; complete one end-to-end flow on Azure first.

1.9 Success Metrics (KPIs)

The following targets will be measured on consented or synthetic test data during Week 8. They are goals, not results.

Area	Metric	Target
Face recognition	Correct identification of registered people (test set)	≥ 90%
Face recognition	False matches (wrong name shown) at the chosen threshold	≤ 2%
Face recognition	Unregistered faces correctly reported as “unknown”	≥ 95%
Face recognition	Time from camera frame to on-screen result	≤ 2 seconds
Reminders	Reminders delivered within 1 minute of the scheduled time (app open)	≥ 95%
Chatbot	Answers supported by an approved record (test questions)	100%
Chatbot	Fallback message shown when the fact is missing	≥ 95%
Usability	System Usability Scale (SUS) score from test users	≥ 70
Project	Integrated MVP deployed on Azure	By Week 7

PART 2

Literature Review

Similar solutions, comparison, and lessons applied to MemoLens

2.1 Background

According to the World Health Organization, around 57 million people were living with dementia worldwide in 2021, with nearly 10 million new cases every year; Alzheimer’s disease may contribute to 60–70% of cases [1]. Dementia organizations recommend memory aids such as calendars, reminders, labelled photos, and consistent daily routines [2][3], and a randomized controlled trial found that smartphone reminder tools improved prospective memory in older adults [4]. These sources support the need for MemoLens; they do not show that MemoLens itself is clinically effective.

2.2 Similar Solutions

Solution	Type	Strengths	Limitations for our users
OrCam MyEye [6]	Wearable camera that attaches to glasses	Works offline; reads text and recognizes saved faces in real time.	Designed for blind and low-vision users, not dementia; high cost; no reminders or personal Q&A.
Envision Glasses [7]	Smart glasses with an AI assistant	Hands-free; describes scenes, reads text, and can recognize faces taught by the user.	Designed for blind and low-vision users; high cost; no caregiver-managed schedule.
Microsoft Seeing AI [8]	Free mobile app	Free; uses the phone camera; describes people and scenes and reads text.	Designed for blind and low-vision users; no reminders, caregiver control, or dementia focus.
Microsoft Project Tokyo [9]	Research prototype on a modified HoloLens	Showed that head-worn cameras can tell a user who is nearby, with audio cues.	Research only; specialized hardware; not available as a product.
Ray-Ban Meta smart glasses [10]	Consumer camera glasses with an AI assistant	Affordable mainstream camera glasses; voice assistant; photo and video capture.	No face recognition of familiar people; general-purpose, not designed for dementia.
Medisafe [11]	Medication reminder app	Reliable medication reminders; caregiver notifications for missed doses.	No face recognition or personal Q&A; requires the user to manage the app.
Voice assistants (Amazon Alexa, Google Assistant)	Smart speakers and phone assistants	Hands-free reminders and general questions.	General web answers can be wrong or confusing; no face recognition; not limited to approved information.

2.3 Comparison with MemoLens

Capability	OrCam	Envision	Seeing AI	Medisafe	MemoLens
Recognizes familiar people	Yes	Yes	Partial	No	Yes
Daily schedule & medication reminders	No	No	No	Yes	Yes
Answers from caregiver-approved information	No	No	No	No	Yes
Caregiver manages the content	No	No	No	Partial	Yes
Designed for people with dementia	No	No	No	No	Yes
Runs on a device the user already owns	No	No	Yes	Yes	Yes

Existing tools each solve one part of the problem. MemoLens combines face recognition, reminders, and a chatbot in one caregiver-controlled system designed for people with dementia, using a phone or laptop the family already owns.

2.4 Technical Approaches Reviewed

- **Face embeddings:** FaceNet maps faces to compact embeddings so that faces of the same person are close together, enabling recognition from a few reference images [5]. ArcFace improves the separation between identities with an additive angular margin loss [12]. MemoLens will compare both in Azure ML and choose based on accuracy and speed.
- **Threshold-based matching:** Comparing embedding distance against a threshold lets the system answer “unknown” instead of forcing a match, which is essential to avoid naming the wrong person.
- **Grounded chatbot answers:** Retrieval-augmented generation supplies the language model with retrieved records so that answers are based on supplied facts rather than the model’s general knowledge [13]. MemoLens applies this idea with caregiver-approved records only.

- **Responsible AI:** Microsoft's Responsible AI principles guide fairness testing, privacy, and transparency [14].

2.5 Lessons Applied to MemoLens

- Keep the experience hands-free and glanceable, as wearable assistants do, by designing the patient view as a glasses display.
- Use existing devices first to keep costs low, and treat dedicated glasses as future work.
- Prefer "I'm not sure" over a wrong name or a wrong answer.
- Put caregivers in control of all personal information, as medication apps do with caregiver notifications.

PART 3

Requirements Gathering

Stakeholders, functional requirements, and quality requirements

3.1 Stakeholders and Their Needs

Stakeholder	Role	Needs
Patient (person living with Alzheimer's)	Primary user	Know who is in front of them; remember appointments and medication; get calm, simple, consistent answers to repeated questions.
Caregiver / family member	Administrator of content	Register familiar people and reference images; manage schedules, reminders, and approved facts; trust that the system will not show wrong information; keep the patient's data private.
Familiar people (relatives, friends)	People being recognized	Give consent before their images are used; be able to have their images deleted.
Healthcare professionals	Indirect stakeholder	The system must not give medical advice or replace professional care.
DEPI evaluators and instructors	Project assessors	Clear use of machine learning and Microsoft Azure, documented requirements, measurable results, and a working demo.
Development team	Builders and maintainers	Clear requirements, modular architecture, and a maintainable codebase in one GitHub repository.

3.2 Functional Requirements

ID	Requirement
Caregiver & account management	
FR-01	The caregiver can sign in securely using Microsoft Entra ID.
FR-02	The caregiver can create, edit, and delete familiar-person profiles (name, relationship, short note).
FR-03	The caregiver can upload consented reference images for each profile and delete them at any time.
FR-04	The caregiver can add, edit, and deactivate approved facts used by the chatbot.
Face recognition (glasses simulation)	
FR-05	The patient view opens the phone or laptop camera after the user grants permission.
FR-06	The system detects faces in the camera feed and sends them to the Azure ML endpoint for matching.
FR-07	When a match is above the threshold, the view shows the person's name and relationship as a glasses-style overlay.
FR-08	When no match is above the threshold, the view shows "I'm not sure who this is" instead of a name.

Daily schedule & reminders	
FR-09	The caregiver can create one-time and recurring events and medication reminders with a time and short message.
FR-10	The patient view shows today's schedule in time order.
FR-11	The system sends a notification with sound at the scheduled time while the app is permitted to notify.
FR-12	The patient can mark a reminder as seen; the system does not claim the medication was taken.
AI chatbot	
FR-13	The patient can type a question in the chatbot.
FR-14	The chatbot answers using only the latest active caregiver-approved records.
FR-15	If no approved record answers the question, the chatbot replies that it does not have that information.
FR-16	The chatbot refuses medical-advice questions and suggests contacting the caregiver or a doctor.

3.3 Quality (Non-Functional) Requirements

ID	Category	Requirement
NFR-01	Performance	Face-recognition result is shown within 2 seconds of the face appearing in the camera view.
NFR-02	Performance	Chatbot replies within 5 seconds for a typical question.
NFR-03	Accuracy	At the chosen threshold, wrong-name results are 2% or fewer on the test set; uncertain cases show "unknown".
NFR-04	Reliability	Reminders are delivered within 1 minute of the scheduled time in tested browser states; known limitations are documented.
NFR-05	Security	Caregiver functions require authentication; the patient view is read-only.
NFR-06	Security	All traffic uses HTTPS; keys and connection strings are stored in Azure Key Vault, never in the repository.
NFR-07	Privacy	Images are stored in private Blob Storage; only minimum data is collected; consent is recorded; data can be deleted on request.
NFR-08	Privacy	Development and testing use synthetic or consented data; no real patient data is stored in GitHub.
NFR-09	Usability	Large readable text, high contrast, at most two taps to reach any patient feature, and calm, consistent wording.
NFR-10	Usability	The patient view works on common phone and laptop browsers at phone screen width.
NFR-11	Responsible AI	Model errors are reviewed with the Azure ML Responsible AI dashboard; chatbot inputs and outputs pass Azure AI Content Safety.
NFR-12	Maintainability	Modules (face recognition, reminders, chatbot, backend, frontend) are separated and deployed through GitHub Actions.
NFR-13	Cost	The system runs within student or free Azure tiers, with budget alerts and idle compute shut down.

3.4 Constraints and Assumptions

- The system runs as a web application on a phone or laptop; no glasses or other hardware are built.
- All cloud services run on Microsoft Azure, as required by the Microsoft Machine Learning Track.
- Users have an internet connection; recognition and the chatbot require access to Azure.
- Browser notification behaviour varies by device; delivery when the app is closed is not guaranteed.

- The project must be completed within the 10-week timeline by a team of six.

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